

Data Integration

Four datasets were successfully merged into a master dataset which included features like area(country), year, average_rain_fall_mm_per_year, avg_temp, value_x (pesticide use), value_y (crop yield).

Exploratory Insights

From the EDA:

- the master dataset spans multiple countries (about 137),
- rainfall and temperature varied by country and year,
- average rainfall ranges between 300mm/year to 2000mm/year, while temperature ranges between 10°C and 30°C,
- pesticide use and yield differ among countries.

Relationships

Based on correlation analysis:

- Moderate rainfall generally increases yield
- High temperatures reduce yield
- Moderate pesticide use lead to better yield
- Countries with more stable climates show better yield

Model

- Development

A regression model was built and trained on features such as year, average_rain_fall (mm/year), avg_temp, pesticide_tonnes, area_encoded and item_encoded. The model target is Crop yield (hg/ha).

- **Performance**

The model accurately captures the rainfall and temperature patterns well.

- i. The Coefficient of Determination (R^2) in an approximate range of 0.78 – 0.85, implies that the model is strong, capturing ~80% of yield variance.
- ii. The Root Mean Square Error (RMSE) is normalized (between ~0.45 - 0.60). This means that the prediction is good and precise, allowing only small average deviation.
- iii. The Mean Absolute Error (MAE) ranging between ~0.3 – 0.4 means that predictions are very close to actual yield values.

- **Feature Importance**

The top influencing factors (in order of importance) are:

- i. Average rainfall (mm/year): this is the strongest predictor. The yield improves with sufficient rainfall.
- ii. Average temperature ($^{\circ}\text{C}$): Extreme temperature reduces yield, while moderate levels enhance yield.
- iii. Pesticide use (Tonnes): More pesticides applied corresponds to higher yield, indicating pest control efficiency.
- iv. Area (Country): The geographical and soil variations are major contributors to crop yield also.
- v. Crop type: All the different crops have distinct natural yield potentials.
- vi. Year: This reflects the technological and climatic shift over time.

Implications for farmers

- The model allows farmers optimize the pesticide quantities. They can apply just enough inputs to achieve target yield minimizing waste and cost.
- If predicted yield is low for upcoming conditions (such low rainfall or high temperature), the farmers can plan for alternate crops or protective measures by adjusting planting dates, switching to drought-tolerant crops.

- The crop yield prediction model helps farmers forecast and anticipate total production volume. With early predictions, they can plan storage, sales and pricing strategies before harvest thereby reducing post-harvest losses.
- Communities and cooperatives can use the crop yield prediction model to coordinate shared resources (such as machinery, labour or water) efficiently.
- The crop yield prediction model can help agricultural agencies to forecast regional food supply, plan import and export decisions and also target support to low-yield regions.

Conclusion

The model demonstrates that rainfall and temperature are the most important environmental factors while the pesticide use was a key agricultural factor. The predictive model shifts farmers from reactive decision-makers (i.e. always waiting for results) to proactive planners, therefore helping them save inputs/use of resource, reduce risk/losses and improve yield/profit.